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NATIONAL ADVISORY COMMITTEE FOR AERONAUTICS

TECHNICAL NOTE 2675

MEASUREMENTS OF FLYING QUALITIES OF AN F-47D-30 AIRPLANE TO DETERMINE LATERAL AND DIRECTIONAL STABILITY AND CONTROL CHARACTERISTICS

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The yaw vane used to measure the angle of sideslip was mounted on a boom 1 chord ahead of the left wing tip. The term, indicated sideslip, used in this paper, is the uncorrected reading given by the yaw vane. The changes in sideslip angle obtained from the yaw vane are believed to be correct but the absolute magnitude of sideslip angle may be in error by a small amount because of lack of symmetry of the yaw vane or because of angularity of flow at the yaw vane. The airspeed was measured by means of a swiveling static head and a shielded total head mounted on a boom 1 chord ahead of the right wing tip. These instrument installations are shown in figure 2. The airspeed installation was calibrated for position error. The position error at low speeds was determined by flying the test airplane in formation with another airplane with an airspeed measuring system which had been calibrated with a trailing airspeed head. At high speeds the position error was determined by the fly-by method as described in reference 2. The term, calibrated airspeed, as used in this paper, may be defined by the following equation:

$$V_c = 45.08 f_o \sqrt{q_c}$$

where

V_c calibrated airspeed in miles per hour; that is, the reading in miles per hour that would be given by a standard Army-Navy airspeed meter if it were connected to a pitot-static system free from position error

f_o standard sea-level compressibility correction factor

q_c pressure differential, in inches of water, between total and static head, corrected for position error

The apparatus used to calibrate the strain gages in the ailerons that measured the aileron hinge moments is shown in figure 6.

RESULTS AND DISCUSSION

Physical Characteristics

The physical characteristics of the rudder, aileron, and elevator control systems as measured on the ground are shown in figures 7 to 12. The variation of the elevator and aileron deflections with stick angle is shown in figure 7 and the friction in these two control systems